

3.75 Medium Intermediate | Co-Polarized IR

Measurement report - 66 windows analyzed (10 s, 1 s step) - window is bad if > 30% of its beats fail

1. Measurement condition

Day / experiment	Day 2 / Exp 1
Skin tone	Medium
Vessel depth	3.75mm
Flow speed	Intermediate
BPM (labeled / measured)	90 / 107
Phantom wavelength	n/a (Experiment 1)

LED channel	IR (940 nm)
Polarization channel	Co-Polarized
Polarization setup	Same
Clamp / phantom	Yes
Hardware channel	C15
Recording	2025-10-23T02-09-29 (trial 1, 75 s)

2. Perfusion index

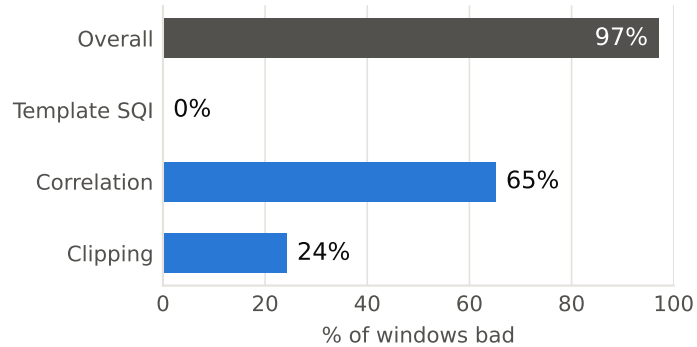
0.50 %

mean $\pm$ 0.05 % SD (across windows)

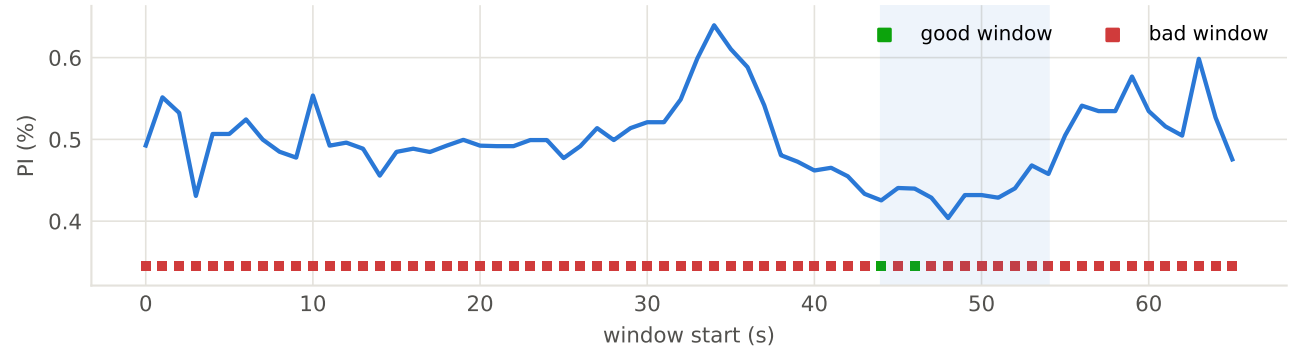
median 0.49 % range 0.40 - 0.64 %

Window PI = median of its beat PIs
(raw signal, AC/DC x 100, DC = beat trough)

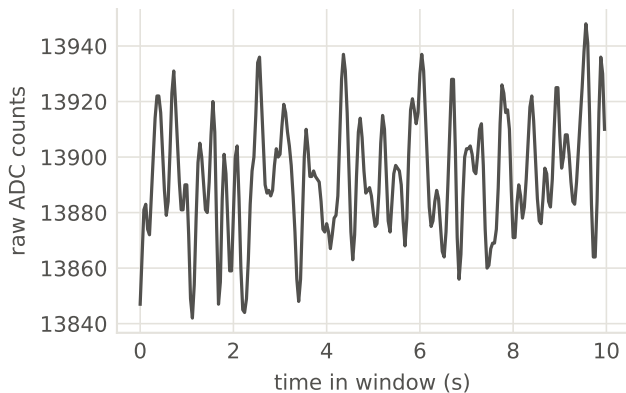
3. Bad window rate



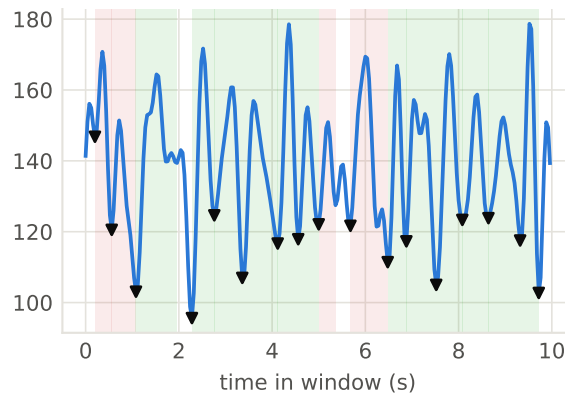
PI per window (strip below = window label; shaded = reference window)



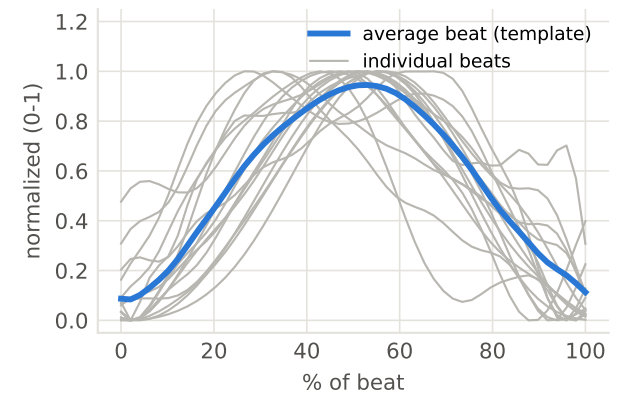
4. Reference window: raw



filtered + beats (12/16 good)



average beat



Reference window: 44-54 s (typical good window; labeled good window). A beat fails if template SQI < 0.05, correlation < 0.8, or clipping SQI < 0.8.

Source: 2025-10-23T02-09-29-d1d3d754-ec8a-4d8e-95d7-606d094f07cf.json